Course-End Project

Agentic RAG: Router-Retriever System with PDF and Web Search Tools

Overview

This project challenges you to design and implement an Agentic Retrieval-Augmented Generation (RAG) system that employs multiple role-based AI agents to process user queries. The workflow includes a Router Agent that classifies questions and a Retriever Agent that executes the most suitable retrieval method, using either a PDF-based vector search or a web search tool. The system demonstrates intelligent orchestration across these agents, producing accurate, source-grounded answers.

Instructions

- Review the learning modules and any linked documentation for CrewAI and the tools used
- Read each section (situation, tasks, actions, and result) to understand the deliverables and workflow expectations thoroughly
- Implement the complete solution using Python in Jupyter notebook, with appropriate orchestration and logging tools
- Submit your final working notebook and a brief README file through the LMS
- Ensure all project elements, from role definitions to reasoning trace visualizations, are included and functional

Situation

As a developer, you are tasked with building an AI-powered multi-agent system to answer domain-specific questions using both static and dynamic sources. The system comprises:

- Router agent: It decides the retrieval path based on the question
- Retriever agent: It executes the retrieval from the chosen source and formulates the answer

Available tools include a PDF search tool for static, domain-specific content and a Web search tool for retrieving fresh information from the internet. An optional generation path directly uses the LLM without retrieval.

Tasks

Your assignment is to build and deliver a multi-agent orchestration system that includes the following:

- Defined roles and responsibilities for at least two agents (router and retriever)
- A coordinated agent workflow that:
    1. Accepts a natural language question as input
    2. Routes the question to the appropriate retrieval method
    3. Executes retrieval and returns a grounded answer
- Interaction logging for reasoning traceability
- A working demo (notebook run) and a brief README file explaining the system's architecture and rationale

Actions

To complete this project, you will have to:

- Define agent roles and capabilities:
    - Router Agent: To analyze the question given as input
    - Retriever Agent: To use the indicated tool to retrieve the output
- Implement tools:
    - PDF search tool using crewai_tools.pdfsearchtool
    - Web search tool using TavilySearchResults
- Select orchestration framework (CrewAI) and justify choice
- Implement agent collaboration with clear input-output flows
- Log reasoning and interactions
- Provide a final demonstration of the working system

Result

Upon completion, you will have to submit:

- A functional Jupyter Notebook demonstrating:
  - Multi-agent coordination, where a user question is processed through routing and retrieval
  - Retrieval from both PDF and web sources
- Trace logs explaining each agent's contributions and interactions
- A README file describing:
  - The system architecture
  - Logic and responsibilities of each agent
  - Coordination flow
  - Challenges faced and trade-offs made

This project strengthens your expertise in designing and deploying Agentic RAG systems that integrate multi-source retrieval and intelligent orchestration, a critical capability in building advanced AI-driven applications.